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WHAT IS IT

A NuBred project is organized into phases. Each phase has a defined purpose, a set of observations, and a gate that must be cleared before advancing. The Variety Manager controls all phase decisions.

WHY IT EXISTS

Varietal development is a multi-year process. Without a defined phase structure, decisions are informal, data is fragmented, and value is impossible to trace. NuBred makes the lifecycle explicit, traceable, and governable.

THE FOUR STANDARD PHASES

	TRIAL	PILOT	LAUNCH	SCALE
Type	Experimental	Experimental	Commercial	Commercial
Objective	Screen many selections with few plants	Validate few selections at semi-commercial scale	First commercial milestone. Royalties activated.	Expand according to contract plan
Scale	Many genotypes, few plants per genotype	Few genotypes, more plants. 0.5–1 ha typical.	First contracted hectares	Full expansion
Forecast	Not required	**Required — mandatory gate**	Required	Required
Royalties	Not active	Not active	PBR + Brand royalties active	PBR + Brand royalties active
Capitolato	Not defined	**Defined by end of Pilot — mandatory**	Applied	Applied
Delta EBITDA	Not active	Baseline being built	Not yet active	Active — royalty calculation

A project can start at any phase. There is no mandatory starting point — a client with an already-tested variety may start directly at Launch.

PHASE GATE LOGIC

Every phase transition requires two things: NuBred verifies gate conditions automatically, then the Variety Manager confirms the advance decision. Both are required. Neither alone is sufficient.

Phase	NuBred Checks Automatically	VM Must Confirm	If Gate Fails

Trial → Pilot	All protocol observations completed	Advance decision per genotype: Promote / Repeat / Discard	Phase stays open. VM notified. No forced action.
Pilot → Launch	All protocol observations completed	Production forecast validated + Capitolato defined + Advance confirmation	Phase stays open. Launch blocked until both gates are cleared.
Launch → Scale	Production forecast validated + Royalty reporting complete	Advance confirmation	Phase stays open. Scale blocked until both gates are cleared.

GENOTYPE DECISION AT PHASE GATE

At every gate, the VM makes an individual decision for each genotype in the phase. There are three options.

Decision	What it means	What NuBred does
☑ **Promote**	Genotype advances to next phase	Creates new phase entry for this genotype. Locks previous phase data.
🔁 Repeat	More data needed. Same phase continues.	Extends current phase. New observation cycle opens.
❌ **Discard**	Genotype removed from active development	Genotype archived with full data history. Not deleted.

A discarded genotype is never deleted from NuBred. Its full observation history is preserved and accessible. This is a governance requirement.

CUSTOM PHASES — HOW THEY WORK

Some clients — particularly those with existing projects or complex multi-territory ecosystems — require phases that do not map to the four standard ones. NuBred supports this through a custom phase service.

Item	Detail
Who requests it	The client — during onboarding or contract analysis
Who builds it	NuBred team — after reviewing the client's contracts and ecosystem structure
What it includes	Custom phase names, custom gate logic, custom parameter sets
When it is delivered	Before the client's first project is created in NuBred
Is it self-service?	No. Custom phases are not a UI feature. They are a NuBred service.

DECIDED — DO NOT REOPEN

- The four standard phases are: Trial, Pilot, Launch, Scale
- A project can start at any phase — no mandatory starting point

- Advancing a phase requires both automatic gate verification AND VM confirmation
- Trial and Pilot gate: observations completed + VM confirms
- Launch and Scale gate: observations completed + forecast validated + VM confirms
- The Capitolato must be defined by end of Pilot — it is a hard gate for Launch
- Discarded genotypes are archived, never deleted
- Custom phases are a NuBred service, not a self-service UI feature

OPEN — TEAM TO IMPLEMENT

- UI design for the phase timeline — how phases and genotype status are shown on the project dashboard
- Notification design — which gate events trigger alerts and to which roles
- UI for the per-genotype Promote / Repeat / Discard decision at gate
- Visual differentiation between experimental phases (green) and commercial phases (blue) throughout the app
- Empty state design for a project with no phases yet configured

DO NOT

- Do not allow phase advance without VM explicit confirmation — automation alone is not enough
- Do not delete discarded genotypes — archive them with full history
- Do not build custom phase creation as a self-service UI feature
- Do not apply royalty logic to Trial or Pilot phases
- Do not block project creation if the client skips a phase — starting at any phase is valid
- Do not treat the Capitolato as optional at the Pilot → Launch gate